

Math F113X: Homework Set 4

Answer the following problems from the Fair Division section:

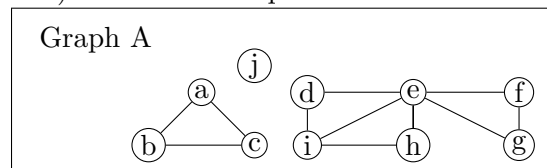
11, 14, 16

Answer the following problems from the Graph Theory section:

A, B, 3, 26, 34a

Note: It will be easier on the grader and on YOU if you do the problems in the order listed.

Problem A: Use the drawing of Graph A (in box) to answer the questions.



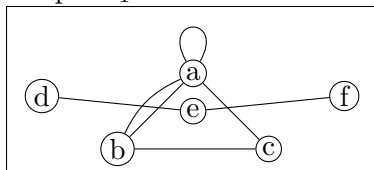
1. How many vertices does Graph A have?
2. How many edges does Graph A have?
3. What is the degree of vertex a? Vertex e? Vertex j?
4. For each sequence of vertices, determine if

it is a **path**, a **circuit**, or **neither**.

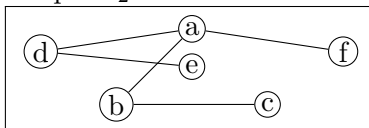
- (a) *abc*
 - (b) *abca*
 - (c) *acid*
 - (d) *id*
 - (e) *gfedie*
 - (f) *gedh*
 - (g) *a*
5. Is Graph A connected? Justify your conclusion.

Problem B: (i) Which of the graphs below are **connected**? (ii) Which contain **circuits**? (iii) Determine the vertices of **highest** degree and those of **lowest** degree. (iv) Determine the number of vertices of **odd** degree.

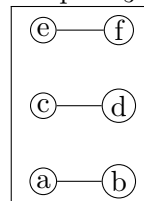
Graph B_1



Graph B_2



Graph B_3



Remember to write up your homework solutions according to the homework writeup guidelines.

Homework is graded using the following rubric for each problem (or problem part):

2 points: You provided a complete answer, with supporting work, written up clearly

1 point: Some attempt at a solution, but incomplete writeup / unclear / illegible / no answer

1 point: Only an answer, with no supporting work

0 points: Missing.

After you do the homework, you need to check your answers against the solutions! Then figure out your errors (if any) and revise your homework before you submit it.